

Degree-Independent Control Impact in the Human Structural Connectome: Graph-Geometric Associations and Degree-Preserving Null Arbitration

Harsha Vardhan Malipeddi*

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Abstract

Question. After accounting for connectivity (degree/strength), which graph-geometric properties explain node-level variation in removal-based control impact (CIS) in the human structural connectome — and do any nodes qualify as operationally defined “hidden chokepoints”?

Methods. 801 QC-pass subjects (AOMIC ID1000; 4S456 parcellation; 15% cost threshold) $\times$ 456 nodes. CIS consumed read-only from the frozen Paper 1 release. The degree-independent residual $R_i = \text{CIS}_i - \widehat{\text{CIS}}(\text{degree}_i)$ is estimated per subject with a frozen cubic-spline family (orthogonality gate: median $|\rho(R, \text{degree})| = 0.0327$). Eleven network features are associated with R within-subject, arbitrated by exact degree-preserving nulls (12 subjects $\times$ 100 nulls, seed 20270927, per-null full CIS + feature recomputation; estimator-equivalence validated), benchmarked out-of-sample (subject-grouped CV), and mapped onto evidence-leveled biological annotations.

Results. (1) The residual is positive in all subjects: $\delta = 0.104$, positive in 801/801 subjects (95% CI [0.1007, 0.1073]). (2) Network position predicts the residual out-of-sample: ridge $\text{cv-}R^2 = 0.574$ (permuted- R control 0.0024; degree-only ≈ 0), strongest associate bridge score (median $\rho = +0.4985$). (3) **Degree-preserving-null arbitration is negative (Outcome C):** every feature’s observed association falls inside the null p95 band — independently deep-verified from the 1,200 raw null records. (4) Joint-criterion “chokepoint” candidates are rare and threshold-sensitive (1/2/3/7 nodes at $k = 1/2/5/10\%$); per-node null arbitration is not established. (5) Top-10%-CIS nodes under-represent cortex (0.444 vs 0.877; $q = 0.00017$) — permutation-supported but not spatially controlled. (6) The fly connectome’s residual also fails its degree-preserving null ($z = 1.21$, $p = 0.109$, frozen): two same-direction negatives at different scales license no universal-mechanism claim.

Conclusion. Degree-independent control impact is a real, replicated, predictable-from-network-features, anatomically non-uniform property of human connectome architecture whose fine structure is **null-qualified**. The “hidden chokepoint” remains an operational, bounded construct rather than a validated node class.

Keywords: structural connectome; network control theory; controllability; degree confounding; null models; graph metrics; cross-species comparison.

*Independent Researcher. Corresponding author: harshavardhanmalipeddi1@gmail.com.

1 Introduction

Network control theory frames the brain as a dynamical system whose structural connectome constrains its reachable dynamics [9, 22, 14]. Removal-based control impact — here **CIS** (Control Impact Score), the fractional loss of average network controllability upon node removal, $CIS_i = (E(G) - E(G - i))/E(G)$ with E the global efficiency [13] — indexes each region’s systemic importance, extending the error/attack-tolerance view of network robustness [2]. Structural connectome construction, streamline weighting and cost thresholding follow established recipes [11, 20, 1].

Paper 1 of this series [18] established two facts that motivate the present work: (i) raw CIS in the human structural connectome is strongly degree-confounded (median $\rho = 0.943$); and (ii) a small but strictly degree-matched residual exists and replicates ($\delta = 0.100$; re-derived here as 0.104). Paper 1 also showed that anatomy does not transfer across scales (“architecture replicates, anatomy doesn’t”): the fly visual system contributes $\approx 80\%$ of total CIS while the human visual system contributes $\approx 2\%$.

Paper 2 asks the mechanistic follow-up, frozen before unblinding (01_CONFIG/CONFIG_FREEZE.md; protocol + Amendments 1–3):

H1 (primary): degree-controlled CIS residuals are associated with specific graph-geometric properties beyond degree/strength.

Secondary hypotheses H2–H5 concern path redundancy, inter-community participation, core/bridge positions, and multivariate explainability; H6–H7 (cross-scale) are exploratory and are never promoted to confirmatory claims. All confirmatory tests are pre-registered against an Outcome A/B/C arbitration rule: **A** = association survives an exact degree-preserving null (mechanism licensed); **B** = mixed; **C** = no association exceeds the null band (mechanism not licensed).

2 Data and node semantics

Human. AOMIC ID1000 derived structural connectomes [21] (Zenodo 19796783, CC-BY-4.0): 801 QC-pass subjects, 4S456 parcellation, SIFT streamline counts [20], 15% cost threshold. CIS values are consumed **read-only** from the frozen Paper 1 release. **Fly (comparison only).** FAFB v783 frozen release artifacts [7, 6]: individual neurons; 3,518 targets. **Node semantics (mandatory).** A human network node is an atlas **parcel** — a macroscopic brain region, never an individual neuron.

3 Methods

3.1 CIS and the degree-independent residual

$CIS_i = (E(G) - E(G - i))/E(G)$, exact recomputation. The residual is $R_i = CIS_i - \mathbb{E}[CIS \mid \text{degree}_i]$ with the conditional expectation estimated **per subject** by OLS of raw CIS on a $df = 4$ natural cubic spline of degree [12]. Gates (all passed): orthogonality median $|\rho(R, \text{degree})| = 0.0327 < 0.05$; same-pairs agreement with Paper 1’s matched-pair Cliff’s δ [5], $\rho = 0.785$. No transform, offset, or retransformation exists in the final estimator (history: Amendments 1–2).

3.2 Feature set and inference

Strength, betweenness [8], closeness, path redundancy, bridge score, participation and within-module z [10], k-core [19], eigenvector centrality [4], PageRank [17], clustering [23] — 11 features + degree. Within-subject Spearman associations (Stage 8); confirmatory within-subject standardized regression with CR1 cluster-robust SEs [15] clustered on subject ($N = 801, 365,256$

rows); 5-fold subject-grouped CV $\times$ 3 seeds for the predictive benchmark (predictive performance is **not** a mechanism claim).

3.3 Degree-preserving null arbitration (confirmatory)

Maslov–Sneppen double-edge rewiring preserving the **exact degree sequence** [16], 12 subjects $\times$ 100 nulls, seed 20270927; per null: full CIS recomputation + full feature recomputation + residualization via the documented 20-bin quantile-median proxy of the frozen spline. The proxy is validated on observed data: the arbitrated association statistic differs between proxy and spline by a median of 0.050 — far below the null bands (0.10–0.73) — and the binned variant yields *stronger* observed associations, so it cannot manufacture Outcome C.

Null ensemble calibration and Monte Carlo precision. The 12 subjects are a frozen-seed random draw from the 801; their pooled statistics are representative (bridge median +0.466 vs +0.4985 population). 100 nulls/subject $\times$ 12 follows the Paper-1 precedent and the compute cost of exact CIS per null. With 1,200 records the p95 rests on the upper ~ 60 order statistics; its Monte-Carlo error is small relative to the band widths (per-feature null IQR of $|\rho| \approx 0.03$ –0.07), and every observed statistic sits strictly inside its band, so the verdict is not boundary-sensitive at this ensemble size.

3.4 Robustness, negative controls, annotation, chokepoint definition, cross-species

Estimator ladder + winsorization (Stage 20); negative controls NC1/NC2 (Stage 21); evidence-leveled annotation + permutation enrichment with BH correction [3] (Stage 23); operational five-criterion chokepoint definition (Stage 22); conservative 12-property human-vs-fly comparison (Stage 22b). Full details: Supplementary Tables S0–S10.

4 Results

4.1 The residual replicates

$\delta = 0.104$ (median, 801 subjects), bootstrap 95% CI [0.1007, 0.1073]; positive in **801/801** subjects; LOSO max deviation 5.4×10^{-8} .

Table 1: Predictive benchmark (5-fold subject-grouped CV $\times$ 3 seeds; numpy-native). Predictive performance is not a mechanism claim.

Model	cv- R^2 (mean $\pm$ sd)	cv- R^2 permuted	Δ vs degree-only
Degree-only	-0.00006 ± 0.000002	0.0004	—
Ridge (11 feat.)	0.5740 ± 0.0001	0.0024	+0.574
Elastic net	0.5614 ± 0.0001	0.0023	+0.561
MLP	-4.09 ± 4.57	−4.65	−4.09 (unstable)

4.2 Associations and controls

All 12 features associate with R at $q < 10^{-27}$ (bridge +0.4985, closeness +0.2489, redundancy −0.2179, participation −0.1676); the 8-feature CR1 model retains all at $q < 10^{-7}$. NC1/NC2 chance bands are $|\rho| \leq 0.0044$; all 8 features exceed both bands (margins large for strong associates, small for the weak ones: within-module z 0.014, k-core 0.052).

Every observed association lies **inside** its null p95 band: the pre-registered **Outcome C**. H1 is therefore **not licensed** by this design.

Table 2: Degree-preserving null arbitration (12 subjects $\times$ 100 nulls; exact degree preservation verified per null; deep-verified by gate V09 from the 1,200 raw records).

Feature	Obs. $ \text{med } \rho $	Null $\text{med } \rho$	Null IQR	Null $ \rho $ p95
Bridge	0.4985	-0.5957	0.0502	0.6538
Clustering	0.1658	0.6846	0.0443	0.7314
Redundancy	0.2179	-0.6228	0.0469	0.6807
Participation	0.1676	0.3909	0.0695	0.4609
Betweenness	0.1333	0.1675	0.0342	0.2102
Within-module z	0.0186	0.0942	0.0448	0.1499
Eigenvector	0.0834	0.0926	0.0317	0.1304
k-core	0.0556	0.0627	0.0323	0.1003
PageRank	0.0379	0.0618	0.0304	0.0961

4.3 Biology (not spatially controlled) and operational chokepoints

Top-10%-CIS nodes under-represent cortex (0.444 vs 0.877; $q = 0.00017$) and over-represent sub-cortex and thalamus; these enrichments are permutation-supported but *not* spatially corrected (parcel centroids unavailable; none invented). Cell-type fields: not available. Chokepoint candidates (CIS top- $k \wedge$ residual top- $k \wedge$ bridge upper half): 1/2/3/7 at $k = 1/2/5/10\%$; node-level null verdict not established.

4.4 Cross-scale comparison

ME.131 (case study only): CIS $z = +23.4$ ($154\times$ peers). The fly population-level residual fails its degree-preserving null ($z = 1.21$, $p = 0.109$; frozen v1.0.0, preserved verbatim). Two same-direction negatives license no universal-law claim.

5 Discussion

The human connectome’s degree-independent control-impact residual is reproducible, positive in all 801 subjects, anatomically biased away from cortex, and substantially predictable from measured network features — yet its feature associations are fully matched by exact degree-preserving nulls. The most defensible reading: what looks like a “chokepoint signature” is largely a consequence of degree architecture plus randomized-topology baselines, and any true fine structure is finer than this design can arbitrate at parcel resolution. This bounds Paper 1 honestly: real, but not yet mechanistically explained.

6 Limitations

(1) Spatial controls not established (no parcel centroids; the enrichment is not spatially corrected). (2) Parcel resolution: no cell-type ground truth. (3) Observational design: no causal claims. (4) 12-subject null ensemble is compute-bound (bands are Monte-Carlo estimates at frozen seed; see calibration paragraph). (5) MLP arm unstable; reported, not interpreted. (6) Fly case study ($n = 1$) does not generalize.

7 Data and code availability

Human: AOMIC ID1000 derived connectomes (Zenodo 19796783, CC-BY-4.0) [21]; atlas: 4S456 frozen labels. Fly: FAFB v783 [7]. Code and artifacts: github.com/harsha-varadhan-2006/humanbrain-hidden tag v2.0.1; the 1,200-record raw null ensemble ships in the release archive (SHA256 in `dist/SHA256SUMS_paper`).

Every stage is scripted with frozen seeds; V01–V12 re-derive all headline numbers independently (12/12 PASS).

8 References

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